

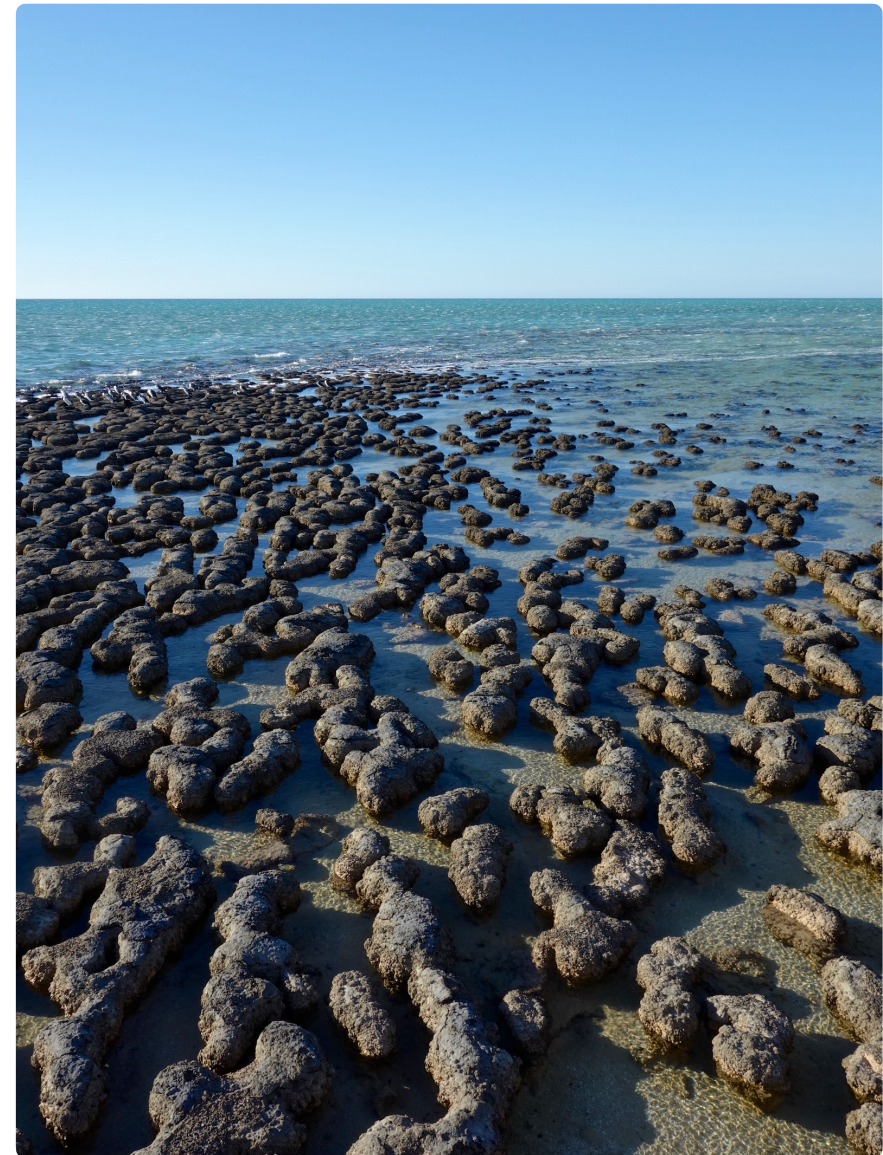
Exercises in Marine Ecological Genetics

03. F -statistics and population structure I

- Calculate F -statistics in R
- Estimate population-specific and pairwise F_{ST}
- Visualize population structure

Martin Helmkamp

<https://github.com/mhelmkampf/meg26>



Today's objectives

By the end of this session, you will be able to:

- Calculate ***F*-statistics** from microsatellite data
- Use F_{ST} to measure **genetic differentiation** (overall, population-specific and pairwise F_{ST})
- Interpret F_{ST} values and **visualize population structure** with Principal coordinates analysis
- Create basic and **improved plots** with tidyverse and ggplot

F-statistics: fixation index

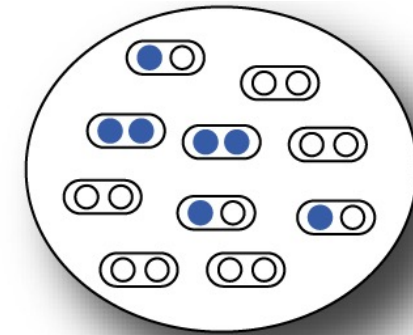
Lecture

HETEROZYGOSITY

In one population

H_o = proportion of heterozygote individuals, observed heterozygosity

$H_e = 2pq = 1 - p^2 - q^2$, expected heterozygosity (assuming HW equilibrium)



$$F = \frac{H_e - H_o}{H_e}$$

Fixation index: *proportion by which heterozygosity is reduced or increased relative to the heterozygosity of a population at HW equilibrium with the same allele frequencies.*

Divided by $H_e \rightarrow$ *proportion* (of expected heterozygosity)

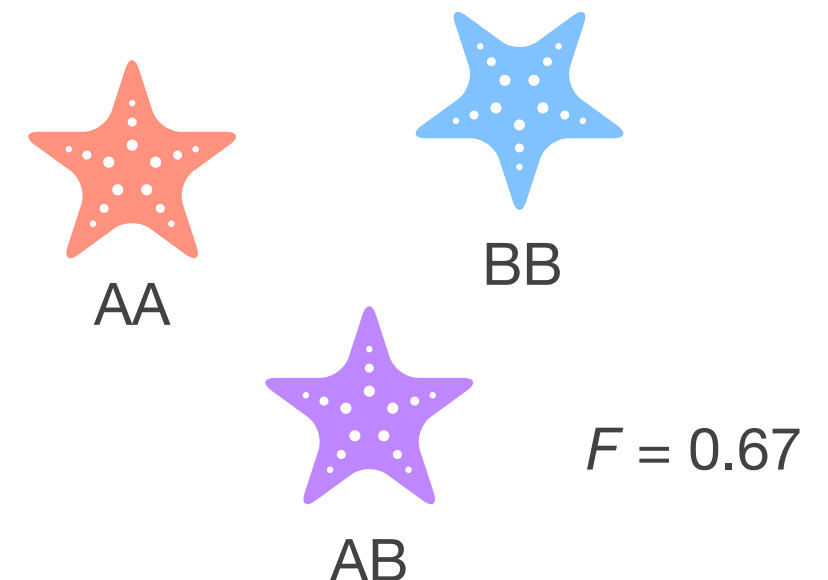
Varies between -1 and 1

$F < 0$: heterozygote excess

$F > 0$ heterozygote deficit (homozygote excess)

May be averaged over several loci $\rightarrow$ reduces bias

May be extended to k alleles



F-statistics: multiple subpopulations

Lecture

In n subpopulations

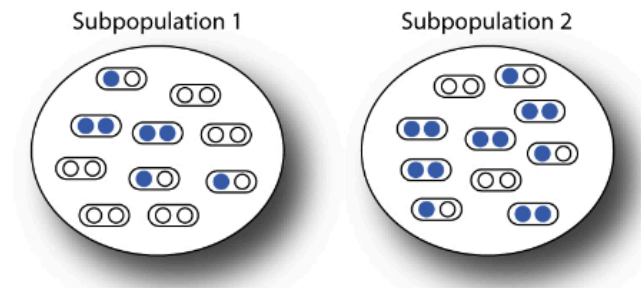
$$H_I = \frac{1}{n} \sum_{i=1}^n \hat{H}_i \quad \text{Mean observed heterozygosity within subpopulations}$$

$$H_S = \frac{1}{n} \sum_{i=1}^n 2p_i q_i \quad \text{Mean expected heterozygosity within subpopulations (assuming HW within subpops)}$$

$$H_T = 2\bar{p}\bar{q} \quad \text{Expected heterozygosity of the total population (assuming HW within the total pop)}$$

$\hat{H}_i$ observed heterozygosity in population i $p_i(q_i)$ frequency of allele A(a) in population i

$\bar{p}(\bar{q})$ mean frequency of allele A(a) n : number of subpopulations



$$F_{IS} = \frac{H_S - H_I}{H_S} \quad F_{ST} = \frac{H_T - H_S}{H_T} \quad F_{IT} = \frac{H_T - H_I}{H_T}$$

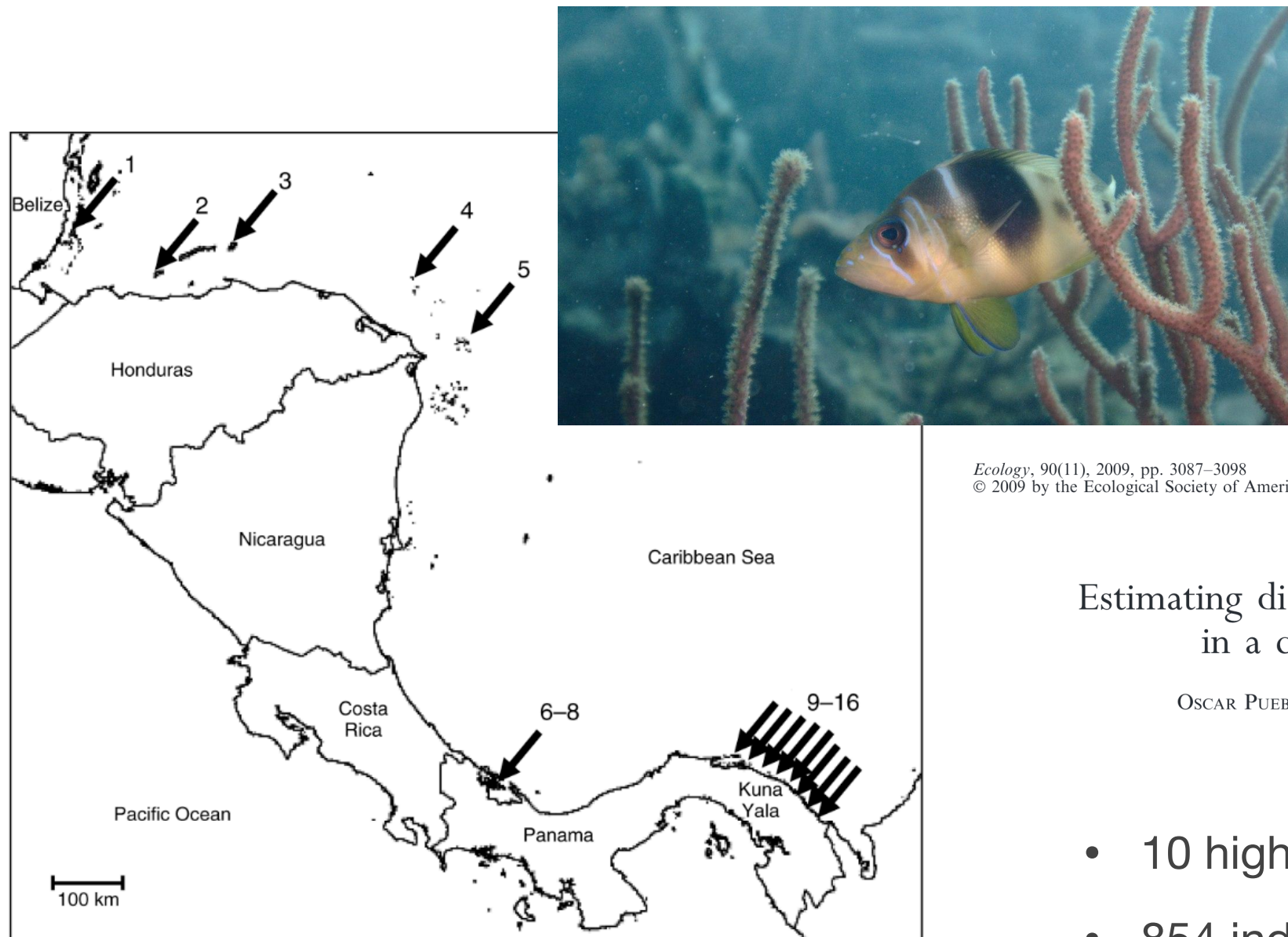
F_{IS} Average difference between observed and Hardy–Weinberg expected heterozygosity within each subpopulation (due to non-random mating)

F_{ST} Reduction in heterozygosity due to subpopulation divergence of allele frequencies

F_{IT} Combined departure from HW expected genotype frequencies due to non-random mating within subpopulations and divergence of allele frequencies among subpopulations

Barred hamlet (*H. puella*) microsatellite dataset

Recap



Estimating dispersal from genetic isolation by distance
in a coral reef fish (*Hypoplectrus puella*)

OSCAR PUEBLA,^{1,2,3} ELDREDGE BERMINGHAM,^{1,2} AND FRÉDÉRIC GUICHARD²

- 10 highly variable microsatellite loci
- 854 individuals
- 15 Caribbean locations

Genepop file format

Recap

puella_barbados.txt

Microsatellite genotypes of Hypoplectrus puella from Barbados

Free text

Locus names

Sample / population

Alleles (000 = missing data)

POP	Sample / population	g2	gag010	h24	hyp001	hyp015	hyp018	e2	hyp008	hyp016	pam013	POP
barbados	ind#735	, 203235	121127	210248	225231	126132	190192	157157	236244	194220	114128	
barbados	ind#736	, 205217	119119	204216	225229	126126	190192	157165	236236	216222	106144	
barbados	ind#737	, 203203	135147	224226	233235	126136	190198	157163	238250	202226	126160	
barbados	ind#738	, 211217	119121	228230	223231	126130	190190	157157	240244	220226	138160	
barbados	ind#739	, 205225	121125	208216	227231	132132	192192	158159	244256	192196	146146	
barbados	ind#740	, 217233	119121	216228	227233	126126	192198	157157	242256	194216	000000	
barbados	ind#741	, 203209	000000	216222	223229	126130	192198	157157	234236	188226	118130	
barbados	ind#742	, 203215	119119	222234	233233	130130	190190	157157	240246	182188	108126	
barbados	ind#743	, 203243	111121	216224	231231	126126	190192	157159	234249	188198	118126	
barbados	ind#744	, 203225	121135	206232	231231	126130	192198	157159	240249	182222	122156	
barbados	ind#745	, 203211	121123	224236	227231	126132	192192	157163	238238	182204	138148	
barbados	ind#746	, 227233	119123	216234	227233	126130	190190	157159	236245	186188	124142	
barbados	ind#747	, 203223	133141	236240	231231	126126	192198	157157	234236	194204	118126	
barbados	ind#748	, 203217	141145	222226	227235	126126	190192	145157	238240	182196	108108	
barbados	ind#749	, 205221	125147	210238	223231	126132	192198	157157	234234	188200	106148	
barbados	ind#750	, 203235	123123	230234	223231	126132	192198	157157	236256	188214	122136	
barbados	ind#751	, 203211	121143	212248	223229	126132	190196	157159	234242	214216	118128	
barbados	ind#752	, 213217	123169	204234	223231	126134	190198	157163	236248	193224	142148	

Connecting to the cluster via SSH

```
### Establish SSH connection to cluster using your course account  
ssh user1234@rosa.hpc.uni-oldenburg.de # please adjust user name
```

```
### Enter password (invisible)
```

...

```
### Either: Download course materials to course account using Git (first time)  
git clone https://github.com/mhelmkamp/meg26.git
```

```
### Or: Update git repository (if it already exists)  
cd meg26  
git pull
```

Keep following instructions in course script (03_popst.R)

F_{ST} range across organisms

Exercise 1

Species	$\hat{F}_{ST}$	$\widehat{N_e m}$	Reference
Amphibians			
<i>Alytes muletansis</i> (Mallorcan midwife toad)	0.12–0.53	1.8–0.2	Kraaijeveld-Smit et al. 2005
Birds			
<i>Gallus gallus</i> (broiler chicken breed)	0.19	1.0	Emara et al. 2002
Mammals			
<i>Capreolus capreolus</i> (roe deer)	0.097–0.146	2.2–1.4	
<i>Homo sapiens</i> (human)	0.03–0.05	7.8–4.6	
<i>Microtus arvalis</i> (common vole)	0.17	1.2	
Plants			
<i>Arabidopsis thaliana</i> (mouse-ear cress)	0.643	0.1	
<i>Oryza officinalis</i> (wild rice)	0.44	0.3	
<i>Phlox drummondii</i> (annual phlox)	0.17	1.2	
<i>Prunus armeniaca</i> (apricot)	0.32	0.5	
Fish			
<i>Morone saxatilis</i> (striped bass)	0.002	11.8	Brown et al. 2005
<i>Sparisoma viride</i> (stoplight parrotfish)	0.019	12.4	Geertjes et al. 2004
Insects			
<i>Drosophila melanogaster</i> (fruit fly)	0.112	2.0	Singh and Rhomberg 1987
<i>Glossina pallidipes</i> (tsetse fly)	0.18	1.1	Ouma et al. 2005
<i>Heliconius charithonia</i> (butterfly)	0.003	79.8	Kronforst and Flemming 2001
Corals			
<i>Seriatopora hystrix</i>	0.089–0.136	2.6–1.6	Maier et al. 200

F_{ST}	Subpopulations are ...
< 0.15	very similar
0.15 – 0.25	similar
> 0.25	distinct

Hamlet species microsatellite dataset

Exercises 1–3

- Complex of 18+ species (genus *Hypoplectrus*)
- Endemic to coral reefs in wider Caribbean
- Little ecological differentiation
- Differ strongly by color pattern
- Reproductively isolated by assortative mating

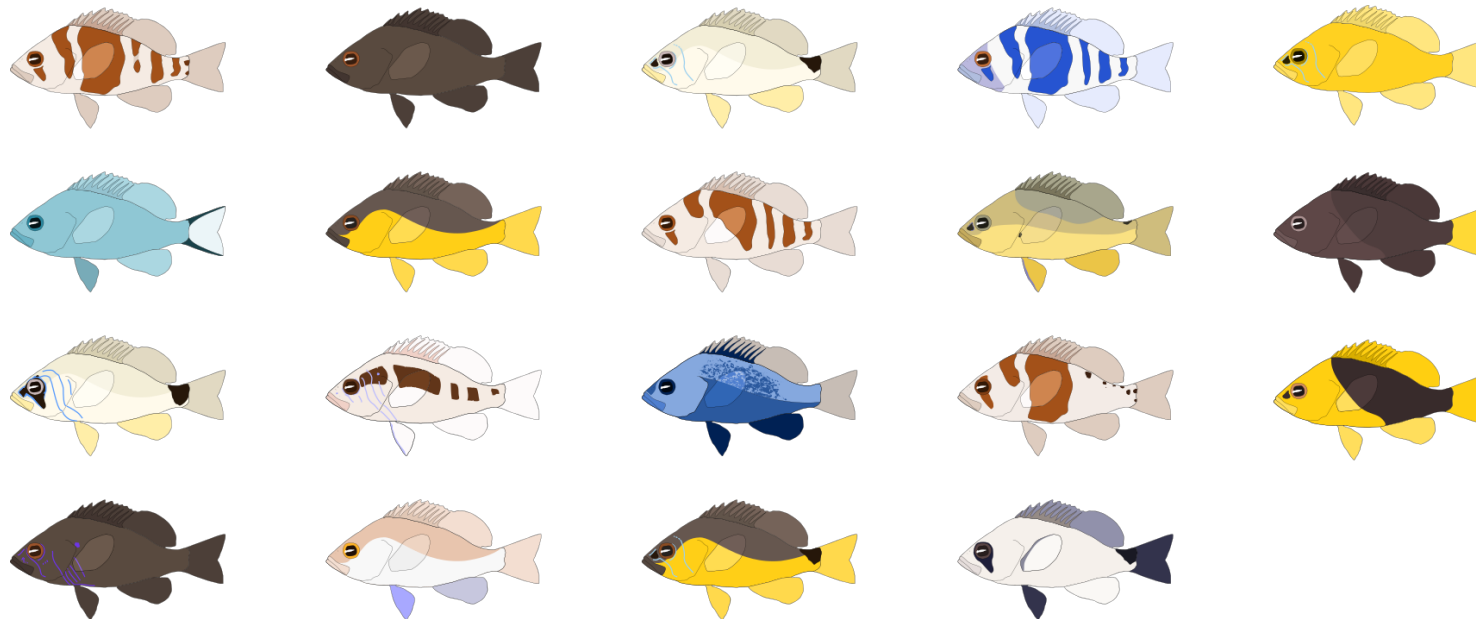
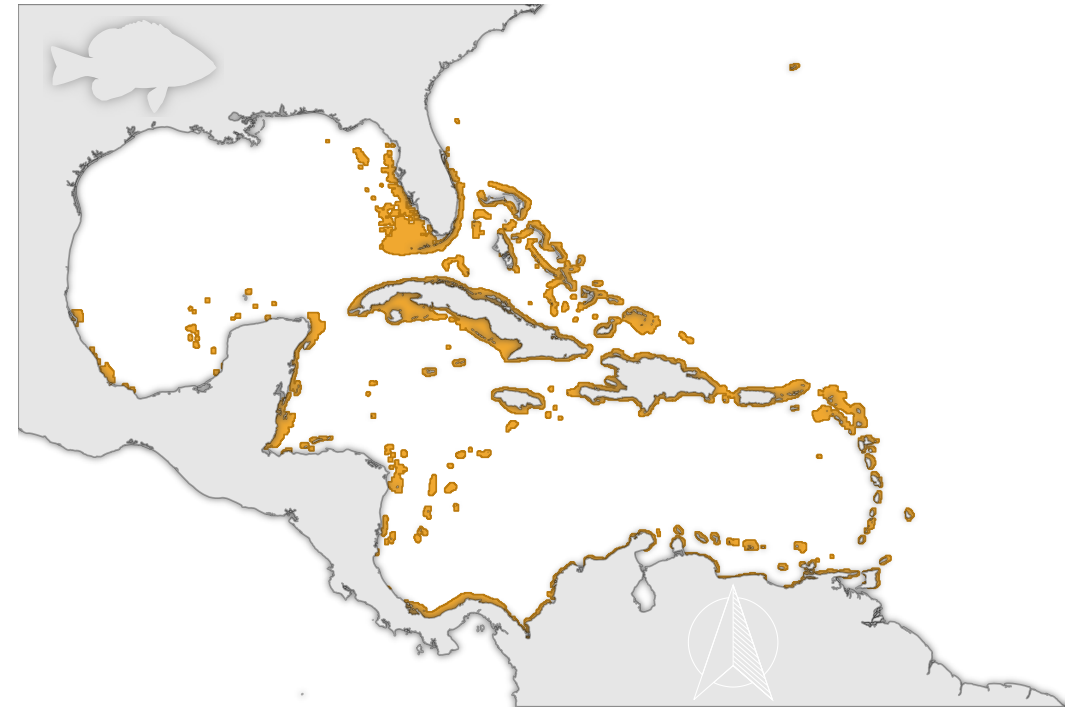


Photo by Allison & Carlos Estape

Illustrations and map by Kosmas Hench, data from biogeodb.stri.si.edu

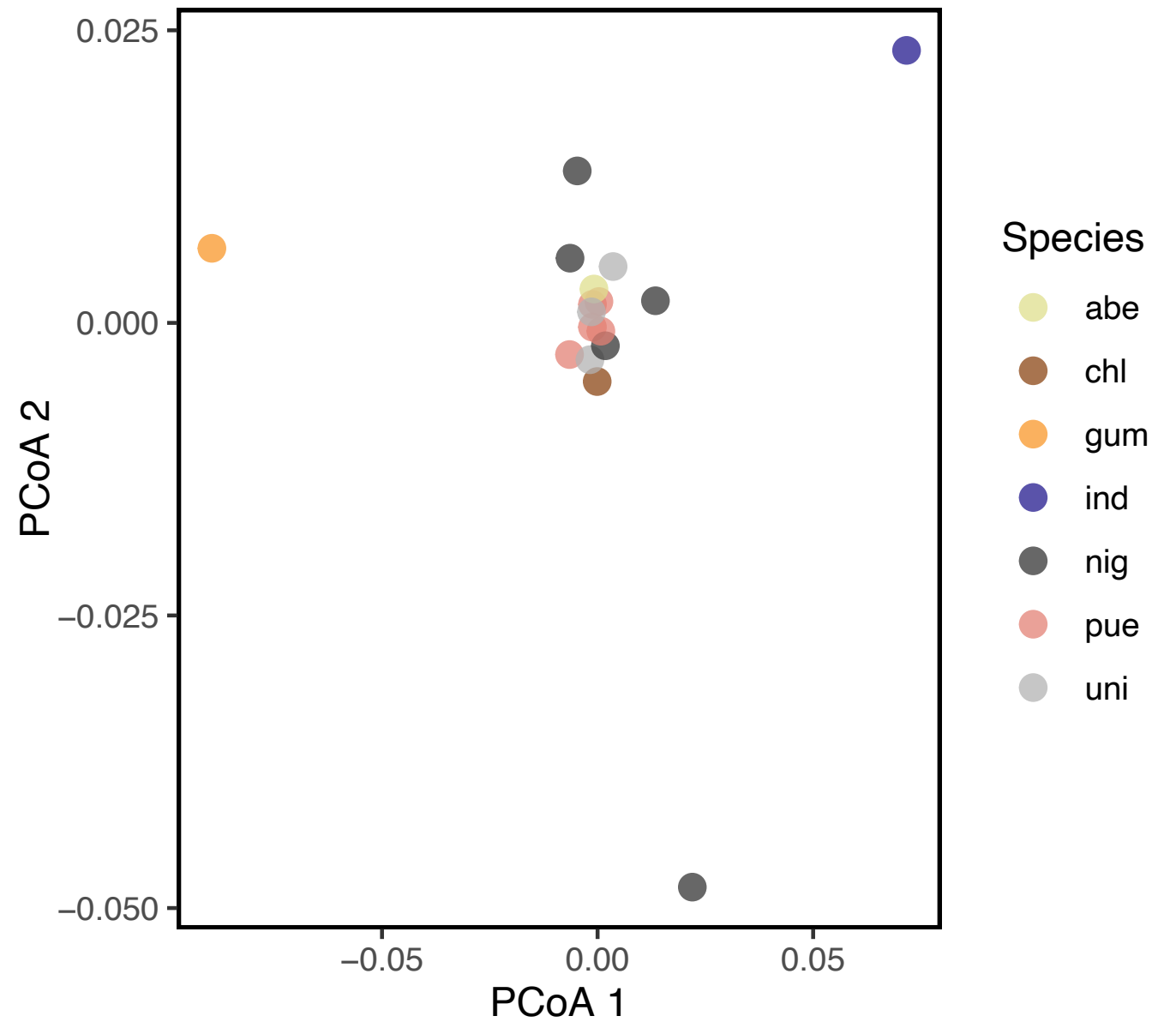
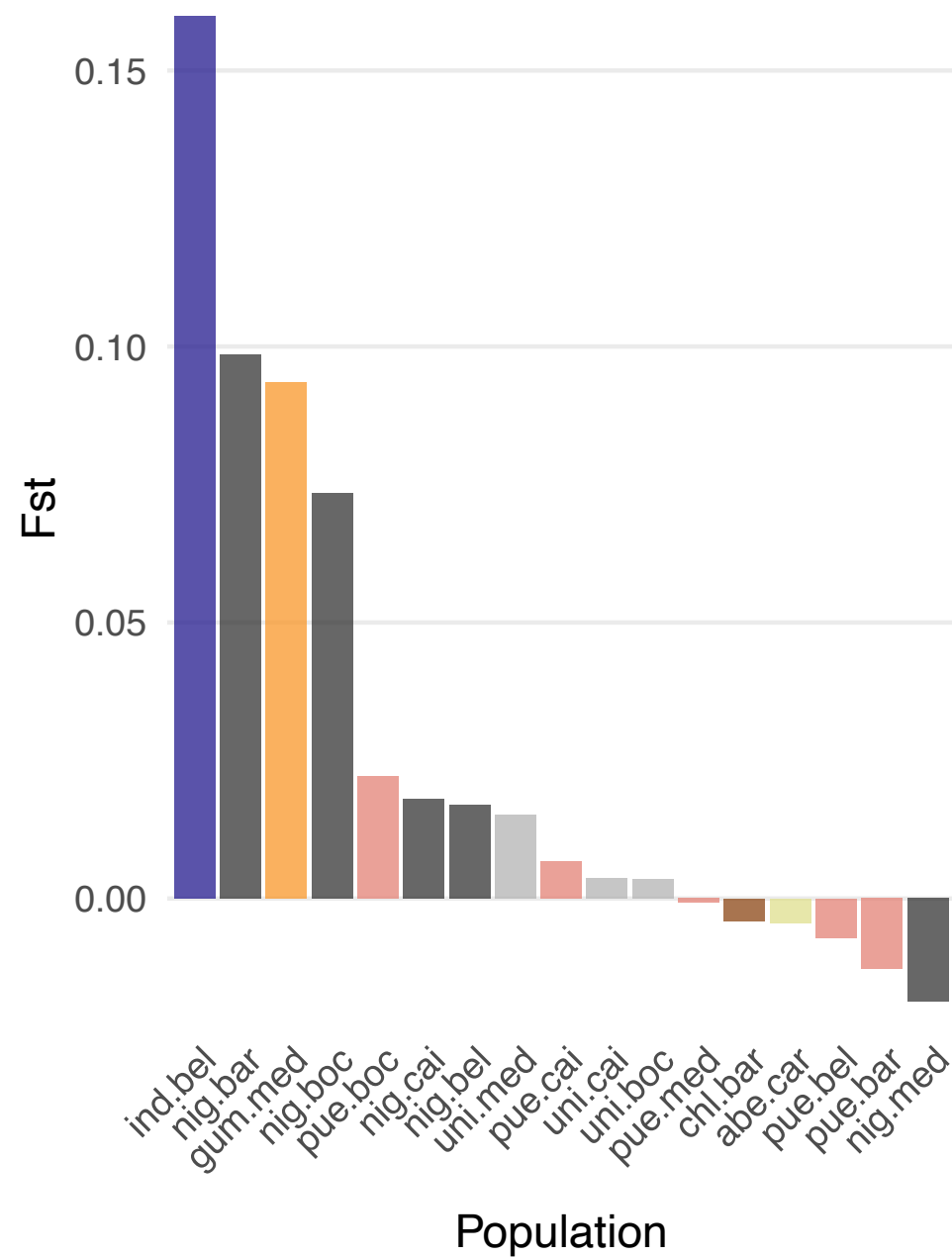
Visualize population structure

Exercise 3

- **Reduce high-dimensional genetic data** to 2–6 axes for visualization
 - Calculate pairwise distances and project samples or subpopulations into a coordinate system that preserves these distances (**PCoA**, principal coordinates analysis)
 - Find axes (principal components) that capture the maximum variance in genotype data (**PCA**, principal component analysis)
- Individuals or subpopulations **cluster by genetic similarity**, revealing structure across populations or species
- PCoA is suited for subpopulations and microsatellite data, while PCA can be used on the level of individuals and SNP genotypes

Visualize population structure

Exercises 2/3



Useful R functions for population genetics

Summary

Calculate F -statistics for all loci

```
pegas::Fst(as.loci(x)) # x = genind object
```

Calculate population-specific F_{ST}

```
betas(x) # hierfstat package
```

Visualize population structure using PCoA and pairwise F_{ST}

```
d <- genet.dist(x, method = "Nei87") # hierfstat package  
p <- pcoa(as.matrix(d))  
# ... plot first two principal components as scatterplot using ggplot
```

New R concepts: tidyverse and ggplot

Summary

Tidyverse pipes

```
tb <- df %>%      # create object (copy)
  mutate() %>%    # 1st operation
  arrange() %>%   # 2nd operation
  select()        # 3rd operation
```

Basic ggplot syntax

```
ggplot(data = tb, aes(x = var1, y = var2)) + # data structure (map variables)
  geom_bar() +                               # geometric shapes representing the data
  labs() +                                   # set plot and axis labels
  guides() +                                 # customize plot legend
  theme()                                    # customize non-data, e.g. fonts, gridlines
```

- **F -statistics** describe how much of the total genetic variation occurs within individuals, within populations, and between populations
- F_{ST} quantifies **genetic differentiation** (how much populations have diverged / degree of gene flow) by comparing within- vs. between-population allele frequency variation
- F_{ST} can be calculated **per population** to assess how distinct each population is, or calculated **pairwise** to visualize population structure (e.g. with PCoA)
- Analyzing **population structure may reveal effects of gene flow, selection, drift**, and informs conservation management

F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to F -statistics.

- How would the number of heterozygotes be affected by self-fertilization?
- What would you expect for the inbreeding coefficient F_{IS} in this case?
- Use the *H. puella* dataset ("caribbean") to check your prediction



Photograph by Luiz Rocha

Course outline

Preliminary, may be subject to change

Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	<i>F</i> -statistics and population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II (optional)	11_edna.R